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Systematic review and meta-analysis of the effects of iodine supplementation on thyroid function and child neurodevelopment in mildly-to-moderately iodine-deficient pregnant women

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ABSTRACT

Background: Mild-to-moderate iodine deficiency, particularly in pregnancy, is prevalent; this is of concern because observational studies have shown negative associations with child neurodevelopment. Although neither the benefits nor the safety of iodine supplementation in pregnancy in areas of mild-to-moderate deficiency are well researched, such supplementation is increasingly being recommended by health authorities in a number of countries.

Objectives: By reviewing the most recent published data on the effects of iodine supplementation in mildly-to-moderately deficient pregnant women on maternal and infant thyroid function and child cognition, we aimed to determine whether the evidence was sufficient to support recommendations in these areas.

Methods: A systematic review of randomized controlled trials (RCTs), non-RCT interventions, and observational studies was conducted. To identify relevant articles, we searched the PubMed and Embase databases. We defined mild-to-moderate iodine deficiency as a baseline median urinary iodine concentration (UIC) of 50–149 μg/L. Eligible studies were included in meta-analyses.

Results: In total, 37 publications were included—10 RCTs, 4 non-RCT interventions, and 23 observational studies. Most studies showed no effect of iodine supplementation on maternal or infant thyroid-stimulating hormone and free thyroxine. Most RCTs found that supplementation reduced maternal thyroglobulin and in 3 RCTs, it prevented or diminished the increase in maternal thyroid volume during pregnancy. Three RCTs addressed child neurodevelopment; only 1 was adequately powered. Meta-analyses of 2 RCTs showed no effect on child cognitive [mean difference (MD): −0.18; 95% CI: −1.22, 0.87], language (MD: 1.28; 95% CI: −0.28, 2.83), or motor scores (MD: 0.28; 95% CI: −1.10, 1.66).

Conclusions: There is insufficient good-quality evidence to support current recommendations for iodine supplementation in pregnancy in areas of mild-to-moderate deficiency. Well-designed RCTs, with child cognitive outcomes, are needed in pregnant women who are moderately deficient (median UIC < 100 μg/L). Maternal intrathyroidal iodine stores should be considered in future trials by including appropriate measures of preconceptional iodine intake. This review was registered at www.crd.york.ac.uk/prospero as CRD42018100277. *Am J Clin Nutr* 2020;112:389–412.

Keywords: iodine, iodine supplementation, pregnancy, mild-to-moderate deficiency, child neurodevelopment, thyroid function, systematic review

Introduction

Although tremendous progress has been made in achieving iodine sufficiency in the general population of many countries (1, 2), some population subgroups, such as pregnant women, still have inadequate iodine intake and are at high risk of

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Abbreviations used: ADHD, attention-deficit hyperactivity disorder; BSID, Bayley Scales of Infant Development; FT_4, free thyroxine; INMA, INfancia y Medio Ambiente; IQ, intelligence quotient; KI, potassium iodide; MD, mean difference; PPTD, postpartum thyroid disease; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RCT, randomized controlled trial; Tg, thyroglobulin; Tg-Ab, thyroglobulin antibody; TPO-Ab, thyroid peroxidase antibody; TSH, thyroid-stimulating hormone; TT_4, total thyroxine; T_4, thyroxine; UIC, urinary iodine concentration.

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deficiency (2, 3). As a result of the physiological changes that occur in pregnancy (4, 5), pregnant women have higher iodine requirements than the general population (6) that can be more difficult to meet through diet, and therefore supplementation may need to be considered. This is of public health concern because iodine deficiency in pregnancy can have adverse consequences on child neurodevelopment (7); however, the role of iodine supplements in pregnancy is not clear.

Iodine is essential for the production of thyroid hormones that are required for brain development in pregnancy and early life. Severe iodine deficiency is known to have a negative effect on thyroid function (8); when it occurs during pregnancy, it may result in cretinism (9). Nowadays, severe iodine deficiency is relatively rare largely as a result of salt-iodization programs (1) but mild-to-moderate iodine deficiency, particularly in pregnant women, is prevalent (2). Observational studies are suggestive of a deleterious effect of maternal mild-to-moderate iodine deficiency on child neurodevelopmental outcomes, e.g., executive function (10), intelligence quotient (IQ) scores (11), reading ability (11), school performance (12, 13), cognitive scores (14), and language skills (15), although not all studies have found significant associations (16–18).

The benefits and safety of iodine supplementation in pregnancy in mild-to-moderate deficiency remain unclear; 2 previous systematic reviews of the effects of iodine supplementation of mildly-to-moderately iodine-deficient mothers found inconsistent results for maternal and infant thyroid function, and that there were few studies with child neurodevelopmental outcomes (19, 20). Despite the insufficient evidence available and the inconsistency of findings, countries are increasingly introducing recommendations for iodine supplementation in pregnancy. The National Health and Medical Research Council in Australia and New Zealand (21) and both the European and American Thyroid Associations (22, 23) recommend supplements containing 150 μg I/d for women who are pregnant or planning a pregnancy. By contrast, the WHO only recommends iodine supplementation in areas with low coverage of iodized salt (24). The United Kingdom has no official recommendations for iodine supplementation (7).

In an attempt to assess whether the current evidence is sufficient to support recommendations for iodine supplementation in pregnancy in areas of mild-to-moderate deficiency, we aimed to review systematically the most up-to-date evidence on the effects of maternal iodine supplementation on maternal and infant thyroid function and child cognition. Our review focuses on summarizing aspects of study design that might explain the inconsistency in results and may need to be considered in future studies (e.g., baseline iodine status and timing of supplementation).

Methods

The reporting of this systematic review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (25). The review is registered with the International prospective register of systematic reviews (PROSPERO: CRD42018100277) and the study protocol is available online (https://www.crd.york.ac.uk/prospero).

Search strategy and eligibility criteria

We searched the PubMed (http://www.ncbi.nlm.nih.gov/pubmed/) and Embase (http://www.embase.com/) databases for relevant articles from inception through to January 2020, using a combination of search terms (**Supplemental Methods**). Additional studies were identified from the reference lists of retrieved publications and by iodine experts.

We included studies on the basis of *1*) the exposure of interest, i.e., iodine suppelementation of any dose, regimen, and type (e.g., iodine-only supplements or iodine as part of multivitamin/mineral tablets or as iodized salt); *2*) the population of interest, i.e., pregnant women with mild-to-moderate iodine deficiency [median urinary iodine concentration (UIC) of 50–149 μg/L (26) at baseline in the total population or in the group of nonusers of iodine supplements and/or iodized salt]; *3*) the outcomes of interest, including maternal and infant/child thyroid function [e.g., thyroid-stimulating hormone (TSH); thyroxine (T_4); thyroglobulin (Tg) concentration; thyroid volume; thyroid-peroxidase antibody (TPO-Ab) titer; thyroglobulin antibodies (Tg-Ab); prevalence of hypothyroxinemia, hypothyroidism, or hyperthyroidism; and any other relevant thyroid effects] and child neurodevelopment (e.g., measures of motor and mental development, IQ scores, language, communication, and behavior-related outcomes); and *4*) study design, i.e., observational, nonrandomized, and/or uncontrolled intervention studies and randomized controlled trials (RCTs); we included all types of study design to maximize information on the effect of iodine supplementation in pregnancy.

Because the focus of this review was on mild-to-moderate iodine deficiency, studies in iodine-sufficient (UIC $\geq$ 150 μg/L) (6) or severely iodine-deficient populations (UIC $<$ 50 μg/L; areas of endemic goiter) (26) were excluded. We excluded supplementation studies in infants or in women with known thyroid disease, studies in languages other than English, unpublished or non-peer-reviewed articles (e.g., meeting abstracts or letters), case-reports, narrative reviews/comment articles, and other systematic reviews or meta-analyses.

Study selection and data extraction

Titles and abstracts of all articles were independently screened by 2 reviewers (MD and HF, or MD and SCB) using a prespecified abstract checklist with the study eligibility criteria. If the abstract did not provide enough information to determine eligibility, the full text was examined; after reviewing all potentially eligible full texts, articles were either excluded (with documented reasons) or included for data extraction. Disagreement between reviewers was resolved through discussion.

Relevant data were independently extracted by 2 reviewers (MD and HF) using a predefined and piloted data-extraction form and included reference details (authors, year of publication), study details (study design, country), participant details (sample size, baseline iodine status), intervention details (type and dosage of supplement, start and duration of intervention, information about placebo/control treatment), details of supplement use for observational studies (study groups and comparisons, iodine dose and vehicle, start and/or duration of reported supplement use), and outcome details (results on all prespecified maternal and infant/child thyroid-function outcomes; time point at measurement; and results on any child-neurodevelopment outcomes, including type of cognitive assessment, assessor, and child age at testing).

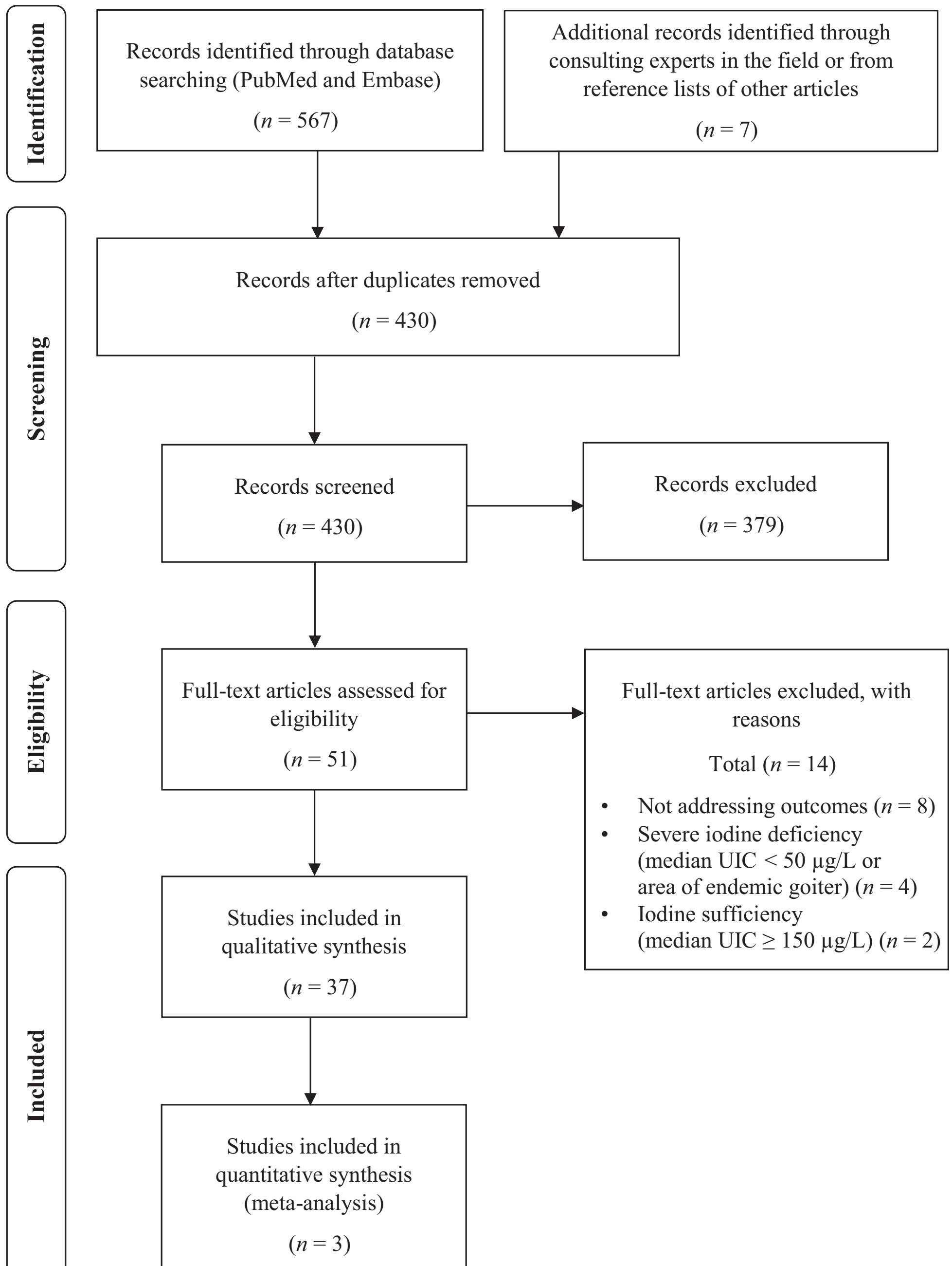

FIGURE 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram of the search results and study selection process. UIC, urinary iodine concentration.

Where possible, the percentage change in maternal thyroid-function parameters during pregnancy was calculated using the first and last reported values; if only 1 measurement was available during gestation, term or postpartum values were used.

Meta-analyses

After assessing the comparability (e.g., design and reported data), the quality, and the risk of bias of the included studies, we performed meta-analyses of RCTs only. Meta-analyses were performed in RStudio version 1.2.5033 (RStudio Inc, Boston, MA, USA). For the RCTs that reported outcome data as medians, the meta-analysis was performed using the pool.med function of the metamedian package developed by McGrath et al. (27). The RCTs that reported outcome data as means were combined using random-effects meta-analysis performed with the metacont function of the meta package.

Risk-of-bias judgment and quality assessment

The Cochrane Risk-of-Bias Tool was used for RCTs and nonrandomized and/or uncontrolled intervention studies (28, 29). Risk of bias was judged as low, high, or unclear for individual elements from 6 domains of bias (selection, performance, detection, attrition, reporting, and other sources of bias). The quality of cohort- and cross-sectional studies was judged using the Newcastle-Ottawa scale (good, fair, or poor) (30) based on 3 study perspectives (i.e., *selection* of study groups, *comparability* of the groups, and *ascertainment* of the outcomes).

Results

The study selection process and the number of studies at each review stage are shown in a PRISMA flow diagram (**Figure 1**). The initial electronic database search yielded a total of 567 records. Seven other potentially relevant articles were identified after consulting experts in the field or the reference lists of articles identified by the search. In total, 144 duplicates were removed and 379 of the remaining records failed to meet the inclusion criteria and were excluded. After full-text review of the remaining 51 records, 37 publications were included, of which 3 were included in the meta-analyses (Figure 1).

Characteristics of included studies

The 37 studies were published between 1981 and 2019; 23 were observational (10 prospective cohort studies and 13 cross-sectional studies) and 14 were interventions (10 RCTs and 4 non-randomized and/or uncontrolled interventions). Only 4 studies addressed all 3 outcomes (maternal and infant thyroid function and child neurodevelopment) (31–34). Studies were mostly based in Europe including Spain (14, 16, 17, 32, 33, 35–38), Italy (39–44), Norway (15, 45–48), Denmark (49–51), France (31, 52), Germany (53, 54), Poland (55, 56), Sweden (57), Latvia (58), and Hungary (59). Four studies were conducted outside Europe [Russia (60), Chile (61), Australia (62, 63), and India and Thailand (34)]. The sample size ranged from 60 to 77,164 in the observational studies and from 35 to 832 in the intervention studies. The forms of iodine supplementation were iodized salt (3 studies) (39, 41, 42), drops of potassium-iodide (KI) solution (2 studies) (50, 61), KI tablets (9 studies) (32–34, 38, 43, 44, 53, 54, 62), and iodine-containing multivitamin/mineral tablets (16 studies) (15, 16, 31, 40, 45–49, 51, 52, 55–59). The exact iodine vehicle was unclear in 5 studies—it was either a KI or an iodine-containing multivitamin/mineral tablet (14, 17, 35, 36, 60); in 2 studies the iodine vehicle was not specified (37, 63). In intervention studies, the dose of iodine ranged from 50 (43) to 300 μg/d (32, 33, 61), mainly as KI tablets. In observational studies, the iodine dose in the supplement-user groups ranged from ≤100 to ≥200 μg/d, mainly in the form of multivitamin/mineral tablets. The exact dose of iodine supplement was not reported or was unclear in 5 studies (36, 37, 41, 47, 56). A few observational studies specifically investigated the effect of initiation of iodine supplements in the preconception period as iodized salt (39–41) or iodine-containing multivitamin/mineral tablets (47, 59).

Risk of bias and quality of included studies

As a result of incomplete reporting, the risk of bias in many of the RCTs and intervention studies was unclear in several domains, particularly those of allocation-concealment and selective-reporting (**Supplemental Figure 1**). Adequate random-allocation sequences were used to generate the study groups in only 5 RCTs (31, 34, 44, 52, 62) and in 1 intervention study (32). Only 3 trials were double-blinded and/or placebo-controlled (34, 44, 51); another study was initiated as double-blind and placebo-controlled but was aborted due to withdrawal of funding support, at which time it was unblinded (62). Most trials and intervention studies had effective blinding of outcome assessors for subjective outcomes (e.g., child neurodevelopment) or the outcomes were objective (e.g., laboratory measurements). The quality of all cohort studies was rated as "good," whereas the cross-sectional studies were mostly of "poor" or "fair" quality (**Supplemental Table 1**).

Maternal thyroid function

The effects of iodine supplements on maternal thyroid function were addressed in 27 studies (13 intervention and 14 observational; **Table 1**).

TSH.

In all 9 RCTs that assessed maternal TSH, there was no difference between the iodine and control groups at various time points during or at the end of the intervention (31, 34, 42, 44, 50–52, 54, 61). In addition, 5 of those trials also found no difference between groups in the change of TSH over time (31, 34, 42, 52, 54). However, 4 RCTs (baseline median UIC in the range of 51–56 μg/L) found a differential effect of iodine supplementation on the trajectory of TSH over the course of pregnancy—2 studies in Denmark (50, 51) reported no change in TSH in the iodine group and an increase in the control group; 1 study in Chile showed a decrease in TSH in the iodine group and no change in the controls (61); and a study in Italy showed a significantly greater increase in TSH in the controls than in the iodine group (44).

A meta-analysis of 2 of the RCTs, which administered 200–225 μg I/d from the first trimester (34, 44), showed significantly lower median TSH in the iodine than in the placebo groups during pregnancy (pooled median difference: −0.08; 95% CI: −0.10, −0.05 mIU/L in the second trimester; −0.34; 95% CI: −0.47, −0.20 mIU/L in the third trimester) (**Supplemental Table 2**).

Three out of 4 non-RCTs found no difference in TSH at different doses (32, 43) or timing of initiation (38) of KI supplementation. Three prospective cohort studies in Italy found lower TSH in pregnant women who used iodized salt for ≥2 y before conception and continued into pregnancy (long-term users) than in those who started using iodized salt on becoming pregnant (39), or in nonusers of iodized salt during pregnancy (41), or in those who took 150 μg I/d from multivitamin/mineral supplements in early pregnancy (40). However, when comparing iodine-supplement users with the nonusers of iodized salt, there was no difference in the proportion with elevated TSH. Results from cross-sectional studies comparing maternal TSH between iodine-supplement users and nonusers during pregnancy were mixed—6 studies found no difference (36, 37, 47, 56, 57, 60), 2 studies found higher (35, 59), and 1 lower TSH (49) in pregnant women who used iodine supplements.

TABLE 1 Summary of findings from 27 studies for the effect of iodine supplements during pregnancy on maternal thyroid function[1]

Study design				Maternal thyroid outcomes				
Study, country	Iodine status[2]	Iodine exposure: dosage,[3] study group (start of supplement) (n)	Iodine vehicles	TSH	FT_4	Serum Tg	Thyroid volume	Other
RCTs ($n = 9$)								
Silva and Silva (61), Chile	54 μg/g	A: 300 μg I/d (9–32 wk) ($n = 36$) B: no iodine ($n = 10$)	KI solution at 785 μg/mL (10 drops)	A: ↓46% B: ↓13% (NS) (9–32 wk vs. 15–40 wk) Decreased in iodine group; no change in no-iodine group; no difference between groups at delivery	Total T_4 was assessed: A: ↑44% B: ↓1% (NS) (9–32 wk vs. 15–40 wk) Increased in iodine group; no change in no-iodine group; higher in iodine group at delivery	N/A	N/A	N/A
Romano et al. (42), Italy	31 μg/24 h	A: 120–180 μg IS/d (≤13 wk) ($n = 17$) B: no IS ($n = 18$)	IS at 20 mg iodide per kg	No change in either group (first to third trimester); no difference between groups at any time point between first and third trimesters	N/A	N/A	A: ↑4% (NS) B: ↑16% (first to third trimesters) Increased in B vs. no change in A; no difference at third trimester between A and B (10.2 vs. 11.7 mL, NS)[4]	N/A
Pedersen et al. (50), Denmark	51 μg/L	A: 200 μg I/d (17–18 wk) ($n = 28$) B: no iodine ($n = 26$)	KI solution (10 drops)	A: ↑5% (NS)[5] B: ↑21% (17 wk vs. 37 wk) Increased in B vs. no change in A; no difference between groups at any time point	A: ↓13%[5] B: ↓20%[5] (17 wk vs. 37 wk) Decreased in both groups with no difference in variation over time or at any time point between groups	A: ↓18%[5] B: ↑60%[5] (17 wk vs. 37 wk) Decreased in A; increased in B; lower in A at all time points	A: ↑16% B: ↑31% (17 wk vs. 37 wk) Increased in both groups with higher increase in the no-iodine group	N/A
Liesenkötter et al. (54), Germany	64 μg/L (53 μg/g)	A: 230 μg I/d (11.2 wk) ($n = 38$) B: no iodine ($n = 70$)	KI tablet (300 μg)	A: ↓32% (NS) B: ↑29% (NS) (10–12 wk vs. PP) No difference between groups at end of intervention PP	Total T_4 was assessed: A: ↓22% (NS) B: ↓40% (NS) (10–12 wk vs. PP) No difference between groups at end of intervention PP	A: ↓50% (NS) B: ↓19% (NS) (10–12 wk vs. PP) No difference between groups PP	A: ↑27% (NS) B: ↑19% (NS) (10–12 wk vs. PP) No difference between groups PP	No change in frequency of TPO-Ab positivity in either group
Nøhr et al. (51), Denmark[6]	51 μg/L	A: 150 μg/d, MV + iodine (P + PP) (11 wk) ($n = 22$) B: 150 μg/d, MV + iodine (P) (11 wk) ($n = 20$) C: MV − iodine ($n = 24$)	MV tablet (+ 50 μg Se)	A and B: ↑4% (NS) C: ↑25% (11 wk vs. 35 wk) Increased in the no-iodine group only; difference in the change over time between groups; no difference between groups at 35 wk	A: ↓5% B: ↓16% C: ↓20% (11 wk vs. 35 wk) Decreased in all groups; no difference in change over time or at 35 wk between groups	A: ↓37% B: ↑5% (NS) C: ↑13% (11 wk vs. 35 wk) Decreased in MV + iodine and increased in MV − iodine; no difference between groups at 35 wk	N/A	No difference in PPTD between groups (59% vs. 60% vs. 46%, NS); no difference in TPO-Ab titers between groups; no difference in percentage with Tg-Ab between groups at 35 wk
Brucker-Davis et al. (52), France	103 μg/L	A: 150 μg/d, MV + iodine (<12 wk) ($n = 32$) B: MV − iodine ($n = 54$)	MV tablet	A: ↓6% (NS) B: ↑16% (NS) (12 wk vs. 33 wk) No change in both groups; no difference between groups at any time point	A: ↓18% B: ↓21% (12 wk vs. 33 wk) Decreased in both groups similarly; no difference between groups at any time point	A: ↓29% B: ↑28% (NS) (12 wk vs. 22 wk) Decreased only in iodine group at second trimester only; lower in iodine group in the second trimester	A: ↓3% (NS) B: ↑8% (NS) (12 wk vs. 33 wk) No change in both groups; no difference between groups at any time point	N/A

(*Continued*)

TABLE 1 (*Continued*)

Study design				Maternal thyroid outcomes				
Study, country	Iodine status[2]	Iodine exposure: dosage,[3] study group (start of supplement) (n)	Iodine vehicles	TSH	FT_4	Serum Tg	Thyroid volume	Other
Brucker-Davis et al. (31), France	117 μg/L	A: 150 μg/d, MV + iodine (<10 wk) ($n = 19$) B: MV − iodine ($n = 25$)	MV tablet	A: ↓3% (NS) B: ↑14% (NS) (12 wk vs. 33 wk) No difference between groups at any time point	A: ↓18% B: ↓17% (12 wk vs. 33 wk) No difference between groups at any time point	A: ↓17% B: ↑61% (12 wk vs. 33 wk) Lower in the iodine group (second and third trimesters)	N/A	N/A
Gowachirapant et al. (34), Thailand and India	131 μg/L	A: 200 μg I/d (10.8 wk) ($n = 412$) B: placebo ($n = 420$)	KI tablet	A: ↑18% B: ↑25% (10–11 wk vs. 30–33 wk) No difference between groups during pregnancy	A: ↓24% B: ↓22% (10–11 wk vs. 30–33 wk) Marginally higher in iodine group during pregnancy	A: ↑5% B: ↑7% (10–11 wk vs. 30–33 wk) Lower in iodine group during pregnancy	A: ↑7% B: ↓1% (10–11 wk vs. 30–33 wk) No difference between groups in pregnancy	No difference in TPO-Ab titers or percentage with raised TPO-Ab (>35 IU/mL) during pregnancy
Censi et al. (44), Italy	56 μg/L (53 μg/g)	A: 225 μg I/d (11 wk) ($n = 52$) B: placebo (10 wk) ($n = 38$)	KI tablet (294 μg)	A: ↑32% B: ↑60% (<12 wk vs. 29–38 wk) Increased in both groups but the increase was significantly greater in B than in A; no difference between groups during pregnancy; lower TSH in A than in B only at 8 wk PP	A: ↓15% B: ↓14% (<12 wk vs. 29–38 wk) Decreased in both groups; no difference between groups at any time point	A: ↓27% B: ↑17% (<12 wk vs. 29–38 wk) Decreased in A; increased in B; lower in A than in B in third trimester only	A: ↑7% B: ↑11% (<12 wk vs. 29–38 wk) Increased in both groups with greater increase in B; no difference between groups at any time point	N/A
Intervention studies ($n = 4$)								
Antonangeli et al. (43), Italy	74 μg/g	A: 200 μg I/d (10–16 wk) ($n = 32$) B: 50 μg I/d, (10–16 wk) ($n = 35$)	KI tablet (262/131 μg)	A: ↑9% (NS)[7] B: ↑9% (NS)[7] (10–16 wk vs. 29–33 wk) No change in either group; no difference between groups at any time point	A: ↓13%[7] B: ↓13%[7] (10–16 wk vs. 29–33 wk) Decreased in both groups similarly; no difference between groups at any time point	A:↓36% (NS)[7] B: ↑25% (NS)[7] (10–16 wk vs. 29–33 wk) NS change in either group; no difference between groups at any time point	A: ↑3% (NS) B: ↑10% (NS) (10–16 wk vs. 29–33 wk) Small increase in B; no difference between groups at any time point in pregnancy; decreased PP in A	No difference in the occurrence of PPTD; no side effects
Berbel et al. (38), Spain[8]	75 μg/L[9]	A: 153 μg I/d (4–6 wk) ($n = 92$) B: 153 μg I/d (12–14 wk) ($n = 102$) C: 153 μg I/d (term) ($n = 151$)	KI tablet (200 μg)	No difference at term between groups A, B, and C (2.10 vs. 2.28 vs. 2.13 mU/L, NS)	Higher in groups A and B (when iodine was started in pregnancy) than in C (at term) (13.3 vs. 13.1 vs. 9.9 pmol/L); no difference between A and B	N/A	N/A	N/A
Velasco et al. (33), Spain	69 μg/L[10]	A: 300 μg I/d (≤13 wk) ($n = 133$) B: no iodine ($n = 61$)	KI tablet (300 μg)	A: ↑8% (NS) B: N/A (first to third trimester) No change in iodine group; lower in iodine group in third trimester (1.99 vs. 2.47 mU/L)	A: ↓26% B: N/A (first to third trimester) Decreased in iodine group; lower in iodine group in third trimester (7.77 vs. 8.98 pmol/L)	A: ↓27% (NS) B: N/A (first to third trimester) Decreased in iodine group but NS	N/A	N/A

(*Continued*)

TABLE 1 (*Continued*)

Study design				Maternal thyroid outcomes				
Study, country	Iodine status[2]	Iodine exposure: dosage,[3] study group (start of supplement) (*n*)	Iodine vehicles	TSH	FT_4	Serum Tg	Thyroid volume	Other
Santiago et al. (32), Spain[11]	109 µg/L	A: 300 µg I/d (<10 wk) ($n = 38$) B: 200 µg I/d (<10 wk) ($n = 55$) C: dose n/r, IS (<10 wk) ($n = 38$)	KI tablet (300/200 µg) and IS	A: ↑43% (NS) B: ↑20% (NS) C: ↑30% (NS) (<10 wk vs. 36 wk) No difference in change over time or at any time point between groups	A: ↓25% B: ↓27% C: ↓28% (<10 wk vs. 36 wk) No difference in change over time or at any time point between groups	A: ↑7% (NS) B: ↓2% (NS) C: ↑0% (NS) (<10 wk vs. 36 wk) No change in all groups; no difference between groups	A: ↑7% B: ↑28% C: ↑3% (<10 wk vs. 36 wk) No difference in change over time or at any time point	N/A
Prospective cohort studies ($n = 3$)								
Moleti et al. (39), Italy	63 µg/L	A: 190 µg/d, long-term IS-user (min 24 mo preconception) ($n = 62$) B: 105 µg/d, short-term IS-user (pregnant) ($n = 38$)	IS at 30 ppm KIO_3	A: ↑28% (first to third trimester) B: ↑37% Lower in long-term IS-users at all time points	A: ↓19% (first to third trimester) B: ↓19% Decreased in both groups; higher in long-term IS-users at all time points	Lower in long-term IS-users: (10.2 vs. 24.1 µg/L)	N/A	82.5% RR reduction of thyroid failure[12] in long- vs. short-term IS-users
Moleti et al. (40), Italy	52 µg/L	A: 150 µg/d, iodine-user (+IS) (9 wk median) ($n = 168$) B: dose n/r, long-term IS-user (min 24 mo preconception) ($n = 105$) C: nonuser ($n = 160$)	MV tablet and IS	A: ↑3% (NS) B: ↑12% (NS) C: ↑7% (NS) (6–9 wk vs. weeks 33–term) Higher in iodine-users than in long-term IS-users at all time points and than in nonusers in late pregnancy; higher percentage with elevated TSH in A than in B and no difference with C	A: ↓21% B: ↓21% C: ↓21% (6–9 wk vs. weeks 33–term) Decreased in all groups; lower in iodine-users than in long-term IS-users and no difference with nonusers; higher percentage with low FT_4 in nonusers than in iodine-users and long-term IS-users	N/A	N/A	N/A
Moleti et al. (41), Italy	48 µg/L	A: dose n/r, IS-user (min 24 mo preconception) ($n = 15$) B: no IS-user ($n = 15$) C: dose n/r, IS-user (+LT_4) ($n = 15$) D: dose n/r, no IS-user (+LT_4) (preconception) ($n = 15$)	IS and LT_4	A: ↑57% B: ↑77% C: ↓34% (NS) D: ↓54% (≤12 wk vs. weeks 31–term) Lower in IS-users than in no IS-users from 13–18 wk onwards	A: ↓23% B: ↓17% C: ↓13% D: ↓7% (≤12 wk vs. weeks 31–term) Higher in IS-users than in no IS-users at all time points; no difference between C and D	N/A	N/A	N/A
Cross-sectional studies ($n = 11$)								
Klett et al. (53), Germany	48 µg/L[9]	A: 135 µg/d, iodine-user ($n = 32$) B: nonuser ($n = 57$)	KI tablet (175 µg)	N/A	N/A	N/A	No difference between A and B at delivery (16.7 vs. 19.5 mL, NS)	N/A
Nøhr and Laurberg (49), Denmark	35 µg/L[9] (39 µg/g)	A: 150 µg/d, iodine-user ($n = 49$) B: nonuser ($n = 95$)	MV tablet	Lower in iodine-users at term (2.06 vs. 2.23 mU/L)	Higher in iodine-users at term (8.4 vs. 7.9 pmol/L)	Lower in A at term (14.7 vs. 25.8 µg/L)	N/A	No difference in frequency of Tg-Ab positivity

(*Continued*)

TABLE 1 (*Continued*)

Study design				Maternal thyroid outcomes				
Study, country	Iodine status[2]	Iodine exposure: dosage,[3] study group (start of supplement) (n)	Iodine vehicles	TSH	FT_4	Serum Tg	Thyroid volume	Other
Fadeyev et al. (60), Russia	44–87 µg/L[13]	A: 150–200 µg/d, iodine-user ($n = 90$) B: nonuser ($n = 125$)	KI or MV tablet	No difference between A and B in any trimester (third trimester: 0.93 vs. 1.26 mU/L, NS)	N/A	N/A	Lower in A in third trimester (12.9 vs. 16.8 mL)	No difference in TPO-Ab positivity
Marco et al. (36), Spain	164 µg/L (135 µg/L in C)	A: dose n/r, iodine-user ($n = 381$) B: dose n/r, IS-user ($n = 75$) C: nonuser ($n = 69$)	KI or MV tablet and IS	No difference between A, B, and C at 26 wk (1.73 vs. 1.67 vs. 2.51 mU/L, NS); no difference in percentage with TSH > 4 mU/L	No difference between A, B, and C at 26 wk (11.6 vs. 11.6 vs. 11.6 pmol/L, NS); no difference in percentage with $FT_4 < 5.1$ pmol/L	N/A	N/A	No difference in Tg-Ab and TPO-Ab positivity and percentage with hypoT_4
Rebagliato et al. (35), Spain	137 µg/L	A: ≥200 µg/d, iodine-user (3 mo preconception up to <24 wk) ($n = 601$) B: 100–199 µg/d, iodine-user (3 mo preconception up to <24 wk) ($n = 298$) C: <100 µg/d, iodine-user/nonuser (3 mo preconception up to <24 wk) ($n = 945$)	KI or MV tablet	Higher percentage with TSH > 3 mU/L in ≥200 µg/d iodine-users (7.5% vs. 6.7% vs. 4.9%); being in ≥200 µg/d iodine-user group vs. <100 µg/d associated with higher TSH and increased risk of TSH > 3 mU/L (OR: 2.5)	Lower in A (≥200 µg/d iodine-users) than in B and C (10.1 vs. 10.9 vs. 10.7 pmol/L)	N/A	N/A	N/A
Menéndez Torre et al. (37), Spain	197 µg/L (138 µg/L in B)	A: dose n/r, iodine-user ($n = 88$) B: nonuser ($n = 85$)	N/A	No difference between A and B (2.30 vs. 1.94 mU/L, NS); higher percentage with TSH > 2.5 mU/L (42.5% vs. 26.5%)	No difference between A and B (15.2 vs. 14.9 pmol/L, NS)	N/A	N/A	No difference in percentage TPO-Ab positive (9.4 vs. 9.1%, NS)
Konrade et al. (58), Latvia	69 µg/L (81 µg/g)	A: ≥150 µg/d, iodine-user ($n = 48$) B: 100–149 µg/d, iodine-user ($n = 70$) C: <100 µg/d, iodine-user/nonuser ($n = 570$)	MV tablet	N/A	Only baseline concentrations in total sample were reported	N/A	N/A	No difference in percentage with TPO-Ab > 60 U/mL between groups (12.8% vs. 8.9% vs. 13.7%) and no difference in odds of elevated TPO-Ab between A and C
Zygmunt et al. (56), Poland	80 µg/L	A: dose n/r, iodine-user ($n = 52$) B: nonuser ($n = 63$)	MV tablets	No difference between A and B (1.76 vs. 1.76 mU/L, NS)	No difference between A and B (13.0 vs. 13.3 pmol/L, NS)	N/A	No difference between A and B (12.2 vs. 11.8 mL, NS)	No difference in TPO-Ab (27.9 vs. 30.6 IU/mL, NS) or Tg-Ab (20.4 vs. 40.2 IU/mL, NS) between A and B
Katko et al. (59), Hungary	162 µg/L (130 µg/L or 113 µg/g in C)	A: ≥150 µg/d, iodine-user (min 4 wk preconception) ($n = 27$) B: ≥150 µg/d, iodine-user (pregnancy ≤ 16 wk) ($n = 51$) C: nonuser ($n = 74$)	MV tablets	Higher in iodine-user (P) than in nonuser (1.97 vs. 1.62 mU/L); no difference between iodine-user (prepregnancy) and nonuser (1.72 vs. 1.62 mU/L, NS)	No difference between A, B, and C (13.9 vs. 13.2 vs. 13.3 pmol/L, NS)	Lower in A than in C (9.1 vs. 14.6 µg/L); no difference between B and C (14.5 vs. 14.6 µg/L, NS)	N/A	N/A

(*Continued*)

TABLE 1 (*Continued*)

Study design				Maternal thyroid outcomes				
Study, country	Iodine status[2]	Iodine exposure: dosage,[3] study group (start of supplement) (n)	Iodine vehicles	TSH	FT_4	Serum Tg	Thyroid volume	Other
Abel et al. (47), Norway	68 μg/L (91 μg/g)	A: dose n/r, iodine-user (1–26 wk preconception) ($n = 347$) B: dose n/r, iodine-user (pregnancy 0–12 wk) ($n = 323$) C: dose n/r, iodine-user (pregnancy 13–20 wk) ($n = 194$) D: nonuser ($n = 1738$)	MV tablet mainly	No difference between groups A, B, C, and D at 18.5 wk (1.2 vs. 1.2 vs. 1.3 vs. 1.2 mU/L, NS)[14]	Lower in C than in D at 18.5 wk (12.4 vs. 12.6 pmol/L, $P = 0.027$); no difference in groups A, B, and D (12.7 vs. 12.6 vs. 12.6 pmol/L, NS)[14]	N/A	N/A	No difference in percentage TPO-Ab or Tg-Ab positive between groups A, B, C, and D (TPO-Ab: 8.1% vs. 9.5% vs. 7.5% vs. 8.5%, NS; Tg-Ab: 8.1% vs. 11% vs. 7% vs. 7.9%, NS)[14]
Manousou et al. (57), Sweden	101 μg/L (110 μg/g)	A: ≥150 μg/d, iodine-user ($n = 253$) B: <150 μg/d, iodine-user/nonuser ($n = 440$)	MV tablet	No difference between groups at median 23 wk (median values per group not reported, $P = 0.63$; overall median measured from DBS 0.7 mU/L)	Total T_4 was assessed: No difference between groups at median 23 wk (median values per group not reported, $P = 0.33$; overall median measured from DBS 124 nmol/L)	Lower in A than in B at 23 wk (19.1 vs. 24.4 μg/L)	N/A	No difference in percentage TPO-Ab positive and percentage with hypothyroidism, hypo-, and hyper-thyroxinemia

[1]DBS, dried blood spot; FT_4, free thyroxine; IS, iodized salt; KI, potassium iodide; LT_4, levothyroxine; MV, multivitamin/mineral supplement; N/A, data not available/reported; n/r, not reported; NS, not statistically significant; P, pregnancy; PP, postpartum; ppm, parts per million; PPTD, postpartum thyroid disease; RCT, randomized controlled trial; Tg, thyroglobulin; Tg-Ab, thyroglobulin antibody; TPO-Ab, thyroid peroxidase antibody; TSH, thyroid-stimulating hormone; T_4, thyroxine; UIC, urinary iodine concentration.

[2]Baseline iodine status was expressed as median UIC (μg/L), median urinary iodine-to-creatinine ratio (μg/g), or median urinary iodine excretion (μg/24 h) of the total sample. If baseline iodine status values were not available for the total sample, values for the control group/group who did not use iodine-containing supplements and/or IS were used.

[3]In some cases where the authors only provided the KI dose used but not the actual iodine content of the tablets, the dose was converted to micrograms of iodide to facilitate comparisons between studies.

[4]Thyroid volume values were estimated from a bar chart in Romano et al. (42).

[5]Values for these parameters were estimated from graphs in Pedersen et al. (50).

[6]Women in this study were a selected group positive for TPO-Ab in Nøhr et al. (51).

[7]The calculated percentage values are based on values for these parameters estimated from graphs in Antonangeli et al. (43).

[8]Women in this study were a very select group (i.e., A = euthyroid compared with B and C = hypothyroxinemic in the first trimester or at term, respectively) in Berbel et al. (38).

[9]Median UIC values at baseline were not reported and values measured at delivery/term were used for these studies: Berbel et al. (38); Klett et al. (53); Nøhr and Laurberg (49).

[10]Baseline iodine status value was based on a subgroup of women from the area who had miscarriage and were not included in this study in Velasco et al. (33).

[11]Women in this study were also split by IS-use for ≥1 y before pregnancy; IS-users had significantly lower thyroid volume in the third trimester than no IS-users (no differences in TSH, FT_4, or Tg were observed), regardless of iodine-supplement use and group allocation in Santiago et al. (32).

[12]Maternal thyroid failure was defined as overt hypothyroidism or isolated hypothyroxinemia during pregnancy in Moleti et al. (39).

[13]Baseline iodine status value was based on the median UIC in parts of the study area, because the median of the exact sample was not reported in this study in Fadeyev et al. (60).

[14]These values were estimated from a figure in Abel et al. (47).

Thyroxine.

Maternal thyroxine concentration was investigated in 8 RCTs; 6 found no difference in free thyroxine (FT_4) (31, 44, 50–52) or total thyroxine (TT_4) (54) between the iodine and control groups, whereas 2 found a slightly higher FT_4 (34) and higher TT_4 (61) in the iodine group. Seven RCTs found that FT_4 or TT_4 decreased during pregnancy and there was no differential effect of iodine supplementation on the change during gestation (31, 34, 44, 50–52, 54).

A meta-analysis of 2 of the RCTs (34, 44) showed no difference in FT_4 between the iodine and placebo groups in the second trimester (pooled median difference: 0.01; 95% CI: −0.12, 0.13 pmol/L) but a significantly lower median FT_4 in the iodine group in the third trimester (pooled median difference: −0.26; 95% CI: −0.38, −0.13 pmol/L) (Supplemental Table 2).

In addition, other intervention studies found either no difference in FT_4 between different doses of iodine supplements (32, 43), higher FT_4 when supplementation was started during pregnancy than at term (38), or lower FT_4 in the iodine group (33). Higher FT_4 in pregnancy was found in long-term users of iodized salt (i.e., started ≥2 y before pregnancy) than in women who either started using it when pregnant (39) or did not use it at all (41). Another study by the same group in Italy found that women who started taking 150 μg I/d (as part of a multivitamin/mineral tablet) had a significantly lower FT_4 during pregnancy than long-term users of iodized salt, although there was no difference compared with nonusers of iodine supplements/salt (40). Results from cross-sectional studies were inconsistent—5 studies found no difference in FT_4 (36, 37, 56, 59) or TT_4 (57), whereas 2 studies found lower (35, 47) and 1 found higher FT_4 (49) in iodine-supplement users.

Thyroglobulin and thyroid volume.

Five of 7 RCTs that measured Tg found it to be lower in the iodine group than in controls (31, 34, 44, 50, 52), whereas there was no difference in 2 trials (51, 54). Four trials showed a decrease in Tg over the course of pregnancy in the iodine group, whereas Tg increased in the controls (31, 44, 50–52).

A pooled meta-analysis of 2 of the RCTs (34, 44) showed that the median Tg during pregnancy was significantly lower in the iodine group than in the placebo group (pooled median difference: −1.1; 95% CI: −1.5, −0.6 μg/L in the second trimester; −2.5; 95% CI: −3.7, −1.2 μg/L in the third trimester) (Supplemental Table 2).

In non-RCT intervention studies, 1 showed no change in Tg during pregnancy in the group taking iodine (33), 1 showed no difference in Tg or its change over time according to different doses of iodine supplement (300 compared with 200 μg/d) (32), whereas another showed a decrease in Tg in response to a higher dose (200 μg/d) compared with an increase with a lower dose (50 μg/d) (43). One cohort study found lower serum Tg in pregnant women who were long-term users of iodized salt than in women who started using it upon becoming pregnant (39). Three cross-sectional studies found lower Tg in women who used iodine supplements than in those who did not (49, 57, 59), with 1 reporting lower Tg only in women who started taking iodine supplements prepregnancy (59).

Five out of 6 RCTs found no difference in thyroid volume between the iodine and the control groups during or at the end of the intervention (34, 42, 44, 52, 54). However, in 3 of the 6 trials, iodine supplementation prevented (42) or diminished the increase in thyroid volume with advancing pregnancy (44, 50). No difference in thyroid volume was reported in 2 intervention studies that used different doses of iodine (32, 43), although 1 of the studies showed a decrease in postpartum thyroid volume only in the group taking the higher dose (200 compared with 50 μg/d) (43). A cross-sectional study found lower thyroid volume in mothers who used iodine supplements than in those who did not (60), whereas 2 studies found no difference between those groups (53, 56).

Thyroid autoimmunity and thyroid disease.

Iodine supplements did not affect thyroid autoimmunity in any of the RCTs; there was no difference in the frequency of TPO-Ab positivity or TPO-Ab titer (34, 51, 54), nor in the percentage with detectable Tg-Ab (51). Furthermore, 1 trial found no difference in the proportion of women that developed postpartum thyroid disease (PPTD), or in its type or severity between the group supplemented with iodine and the controls (51). A later intervention study also reported no enhancement of the occurrence of PPTD or any side effects in the mothers in relation to iodine supplementation (43). None of the 8 cross-sectional studies that investigated the effect of iodine-supplement use compared with no use on markers of thyroid autoimmunity found a difference in the frequency of Tg-Ab (36, 47, 49) or TPO-Ab positivity (36, 37, 47, 57, 58, 60), or their titers (56). Risk of maternal thyroid failure (overt hypothyroidism or isolated hypothyroxinemia in pregnancy) was reduced with long-term (started prepregnancy) but not short-term (started in pregnancy) use of iodized salt, as reported in an observational cohort study (39).

Infant/child thyroid function

Markers of infant/child thyroid function were assessed in 14 studies (9 intervention and 5 observational) (**Table 2**).

Cord-blood markers.

The majority of studies showed no effect of maternal iodine supplementation on cord-blood TSH or FT_4. None of the 4 RCTs that reported cord-blood TSH showed any difference between the iodine and control groups (31, 50, 61, 62). One of 2 non-RCT intervention studies also showed no effect of different doses of maternal iodine supplementation on cord-blood TSH (32). Only 2 studies—1 cross-sectional (49) and 1 inadequately controlled intervention (33)—showed an increase in cord-blood TSH in the iodine-supplemented women. Three of 4 RCTs that assessed cord-blood FT_4 or TT_4 showed no difference between the iodine and control groups (31, 50, 62). Only 1 of the RCTs showed higher cord-blood TT_4 in the iodine group than in the controls (61). Only 1 cross-sectional study reported on FT_4; it showed higher cord-blood FT_4 in mothers who used iodine than in nonusers (49). One RCT reported lower cord-blood Tg in the intervention than in the control group (50), whereas 2

TABLE 2 Summary of findings from 14 studies for the effect of iodine supplements during pregnancy on neonatal/child thyroid function[1]

Study design				Neonatal/child thyroid outcomes				
Study, country	Iodine status[2]	Iodine exposure: dose,[3] study group (start of supplement) (n)	Iodine vehicles	TSH	FT_4	Serum Tg	Thyroid volume	Other
RCTs ($n = 7$)								
Silva and Silva (61), Chile	54 μg/g	A: 300 μg I/d (9–32 wk) ($n = 36$) B: no iodine ($n = 10$)	KI solution at 785 μg/mL (10 drops)	No difference between groups A and B in cord-blood (5.7 vs. 8.3 mU/L, NS)	Higher total T_4 in iodine group (145.4 vs. 119.7 nmol/L)	N/A	N/A	N/A
Pedersen et al. (50), Denmark	51 μg/L	A: 200 μg I/d (17–18 wk) ($n = 28$) B: no iodine ($n = 26$)	KI solution (10 drops)	No difference between groups A and B in cord-blood (6.8 and 7.8 mU/L, NS)	No difference between groups A and B (13.6 and 13.6 pmol/L, NS)	Lower in iodine group (38 vs. 67 μg/L)	N/A	N/A
Liesenkötter et al. (54), Germany	64 μg/L (53 μg/g)	A: 230 μg I/d (11.2 wk) ($n = 38$) B: no iodine ($n = 70$)	KI tablet (300 μg)	Measured in newborns but not reported	N/A	N/A	Lower in iodine group (0.7 vs. 1.5 mL)	No difference in frequency of TPO-Ab placental transfer between groups
Brucker-Davis et al. (31), France	117 μg/L	A: 150 μg/d, MV + iodine (<10 wk) ($n = 19$) B: MV − iodine ($n = 25$)	MV tablet	No difference between groups A and B in cord-blood (8.0 vs. 6.2 mU/L, NS)	No difference between groups A and B (13.4 vs. 12.8 pmol/L, NS)	No difference between groups A and B (66.8 and 96.1 μg/L, NS)	N/A	N/A
Zhou et al. (62), Australia	137 μg/L	A: 150 μg I/d (15.2 wk) ($n = 29$)[4] B: placebo ($n = 30$)[4]	KI tablet	No difference between groups A and B in cord-blood (8.2 vs. 6.6 mU/L, NS) or in newborns (2.1 vs. 2.2 mU/L, NS)	No difference between groups A and B (14.4 vs. 13.8 pmol/L, NS)	No difference between groups A and B (73 vs. 64 μg/L, NS)	N/A	N/A
Gowachirapant et al. (34), Thailand and India	131 μg/L	A: 200 μg I/d (10.8 wk) ($n = 412$)[5] B: placebo ($n = 420$)[5]	KI tablet	No difference between groups at birth, 1 y, 2 y, or 5 y	No difference between groups in total T_4 at birth, 1 y, 2 y, or 5 y; FT_4 measured in newborns and at 2 y but not reported	N/A	Measured at 2 y but not reported	N/A
Censi et al. (44), Italy	56 μg/L (53 μg/g)	A: 225 μg I/d (11 wk) ($n = 52$) B: placebo (10 wk) ($n = 38$)	KI tablet (294 μg)	No difference between groups A and B in newborns (2.7 vs. 3.6 mU/L, NS)	N/A	N/A	N/A	N/A

(*Continued*)

TABLE 2 (*Continued*)

Study design				Neonatal/child thyroid outcomes				
Study, country	Iodine status[2]	Iodine exposure: dose,[3] study group (start of supplement) (n)	Iodine vehicles	TSH	FT_4	Serum Tg	Thyroid volume	Other
Intervention studies ($n = 2$)								
Velasco et al. (33), Spain	69 μg/L[6]	A: 300 μg I/d (≤13 wk) ($n = 133$) B: no iodine ($n = 61$)	KI tablet (300 μg)	Higher in iodine group in cord-blood (7.93 vs. 3.77 mU/L)	N/A	N/A	N/A	N/A
Santiago et al. (32), Spain	109 μg/L	A: 300 μg I/d (<10 wk) ($n = 38$) B: 200 μg I/d (<10 wk) ($n = 55$) C: dose n/r, IS (<10 wk) ($n = 38$)	KI tablet (300/200 μg) and IS	No difference between groups A, B, and C in cord-blood (3.22 vs. 2.49 vs. 2.98 mU/L, NS)	N/A	N/A	No difference between groups A, B, and C (0.42 vs. 0.42 vs. 0.49 mL, NS)	N/A
Cross-sectional studies ($n = 5$)								
Klett et al. (53), Germany	48 μg/L[7]	A: 135 μg/d, iodine-user ($n = 32$) B: nonuser ($n = 57$)	KI tablet (175 μg)	No difference between groups in newborns (exact values not reported by group)	N/A	N/A	Lower in iodine group (1.0 vs. 1.2 mL)	N/A
Nøhr and Laurberg (49), Denmark	35 μg/L[7] (39 μg/g)	A: 150 μg/d, iodine-user ($n = 49$) B: nonuser ($n = 95$)	MV tablet	Higher in iodine group in cord-blood (9.00 vs. 7.07 mU/L)	Higher in iodine group (12.5 vs. 11.7 pmol/L)	Lower in iodine group (34.3 vs. 56.7 μg/L)	N/A	No difference in frequency of Tg-Ab positivity between groups
Gietka-Czernel et al. (55), Poland	113 μg/L	A: 150 μg/d, iodine-user ($n = 35$) B: nonuser ($n = 65$)[8]	MV tablet	No difference between groups A and B in newborns (1.57 vs. 1.33 mU/L, NS)[9]	N/A	N/A	N/A	N/A

(*Continued*)

TABLE 2 (*Continued*)

Study design				Neonatal/child thyroid outcomes				
Study, country	Iodine status[2]	Iodine exposure: dose,[3] study group (start of supplement) (n)	Iodine vehicles	TSH	FT_4	Serum Tg	Thyroid volume	Other
Marco et al. (36), Spain	164 μg/L (135 μg/L in C)	A: dose n/r, iodine-user ($n = 381$) B: dose n/r, IS-user ($n = 75$) C: nonuser ($n = 69$)	KI or MV tablet and IS	No difference between groups in newborns (exact values not reported by group)	N/A	N/A	N/A	N/A
Mitchell et al. (63), Australia	96 μg/L[10]	A: 200–300 μg/d, iodine-user ($n = 43$) B: 150 μg/d, iodine-user ($n = 21$) C: 0–75 μg/d, iodine-user/nonuser ($n = 47$)	N/A	No difference between groups A, B, and C in newborns (2.0 vs. 2.2 vs. 2.0 mU/L, NS)[11]	N/A	N/A	N/A	N/A

[1]FT_4, free thyroxine; IS, iodized salt; KI, potassium iodide; MV, multivitamin/mineral supplement; N/A, data not available/reported; n/r, not reported; NS, not statistically significant; RCT, randomized controlled trial; Tg, thyroglobulin; Tg-Ab, thyroglobulin antibody; TPO-Ab, thyroid peroxidase antibody; TSH, thyroid-stimulating hormone; T_4, thyroxine; UIC, urinary iodine concentration.

[2]Baseline iodine status was expressed as median UIC (μg/L), median urinary iodine-to-creatinine ratio (μg/g), or median urinary iodine excretion (μg/24 h) of the total sample. If baseline iodine status values were not available for the total sample, values for the control group/group who did not use iodine-containing supplements and/or IS were used.

[3]In some cases where the authors only provided the KI dose used but not the actual iodine content of the tablets, the dose was converted to micrograms of iodide to facilitate comparisons between studies.

[4]Numbers for the cord-blood tests were smaller (19 and 22 for group A and B, respectively) in Zhou et al. (62).

[5]Numbers for child blood tests were smaller (e.g., at 5 y there were 159 in each group) in Gowachirapant et al. (34).

[6]Baseline iodine status value was based on a subgroup of women from the area who had miscarriage and were not included in this study in Velasco et al. (33).

[7]Median UIC values at baseline were not reported and values measured at delivery were used for these studies: Klett et al. (53); Nøhr and Laurberg (49).

[8]Some of the women in group B ($n = 35$) also took MV tablets (without iodine) in Gietka-Czernel et al. (55).

[9]TSH measurements were performed for a total of 68 newborns and numbers per group were not reported in Gietka-Czernel et al. (55).

[10]Baseline iodine status value was based on the median UIC in a previous study of pregnant women from the region [Rahman et al. (64)], because the baseline iodine status of the exact sample was not reported in Mitchell et al. (63).

[11]TSH values for groups A and C were estimated from a bar chart in Mitchell et al. (63).

reported no difference between the groups (31, 62). Only 1 cross-sectional study reported on serum Tg and found lower cord-blood concentration in iodine-users than in nonusers (49). The same study also found no difference in the frequency of cord-blood Tg-Ab positivity between iodine-users and nonusers (48).

Thyroid function in the infant/child.

None of the 3 RCTs that reported TSH in newborns (34, 44, 62) or in children (34) found a difference between the iodine and control groups. There was also no difference in infant TSH between supplement users and nonusers in all 4 cross-sectional studies (36, 53, 55, 63). Only 1 of the RCTs reported on FT_4 and TT_4 in newborns and in children; it showed no difference between the iodine and control groups (34). The only RCT that reported on infant thyroid volume found that it was lower in the iodine group than in the control group (54). One uncontrolled intervention study found no difference in infant thyroid volume between 2 groups of mothers taking different doses of iodine and a group using only iodized salt (32). A cross-sectional study found lower thyroid volume in infants of mothers who were iodine-supplement users than in those of nonusers (53). There was no increase in the frequency of TPO-Ab placental transfer in infants of iodine-supplemented mothers in an RCT (54).

Child neurodevelopment

Child neurodevelopmental outcomes were reported in 14 studies (6 intervention and 8 observational) (**Table 3**). Three RCTs (baseline median UIC: 117–137 μg/L) that gave doses of 150 or 200 μg I/d as KI or iodine-containing multivitamin/mineral tablets studied the effect of iodine supplementation in pregnancy on child neurodevelopment (31, 34, 62); however, only 1 was adequately powered (34). We performed meta-analyses of the 2 RCTs that were comparable in study design [e.g., both administered iodine as KI tablets in a similar dosage (150 or 200 μg I/d) from early pregnancy until delivery, used a placebo tablet as the control treatment, assessed child cognition using the same cognitive tool (Bayley scales), and children were tested at a similar age (at 1.5 or 2 y)] (34, 62). We did not include the other RCT because it used iodine as part of multivitamin/mineral tablets (31). The meta-analyses showed that there was no effect of iodine supplementation compared with placebo on child cognitive, language, or motor scores (**Figure 2**).

Motor development.

None of the 3 RCTs found either a beneficial or harmful effect of maternal iodine supplementation on child motor development measured at 1–2 y (31, 34, 62). Three nonrandomized and/or inadequately controlled intervention studies looked at the effect of iodine supplementation on child neurodevelopment (32, 33, 38) and in 2, there was a beneficial effect of iodine supplementation, given as 200 or 300 μg/d as KI, on motor development at 1.5 y (33, 38). Maternal iodine-supplement use in dosages ≥150 μg/d compared with 0–99 μg/d (taken as iodine-containing multivitamin/mineral or KI tablets) was negatively associated with child psychomotor development at 1 y when estimates were pooled from 4 subcohorts of the Spanish INMA (INfancia y Medio Ambiente) prospective multicenter study (16, 17); however, the overall estimate was driven by only 2 of the subcohorts (i.e., Valencia and Asturias). In a later study by the same group, negative effects on motor development were not confirmed at age 4–5 y in any of the subcohorts nor when the estimates were pooled (14). Two cohort studies in Norway showed mixed results: 1 found no association between maternal iodine-supplement use and child fine or gross motor development at 3 y, nor with the odds of not walking unaided at 17 mo (45); whereas the other study showed a negative association with gross motor but not with fine motor development in infancy and toddlerhood (15).

Mental development.

Cognitive development and language at age 1–2 y, IQ, processing speed, global executive function, and auditory performance at 5.4 y did not differ between the iodine and the control groups in the RCTs (31, 34, 62). In the non-RCT intervention studies, there was no beneficial effect on mental development (32, 33), language quotient (38), or total development (32, 38). Observational studies also did not find associations between supplement use and overall mental development scores at 1 y (16, 17), cognitive and language scores at 6 and 12 mo (15), language or communication delay at 3 y (45), cognitive function at 4–5 y (14), or child language, reading, and writing skills at 8 y (48). By contrast, a cohort study in Italy reported a beneficial effect of iodine, supplied as iodized salt for ≥2 y preconception (long-term use), on child IQ at 6–12 y (41).

Behavioral development.

There was no effect of iodine supplementation in RCTs on child behavior, including adaptive behaviors at 1.5 y (62), social-emotional behaviors at 1.5 (62) and 2 y (31), and total difficulties at 5.4 y (34). Two non-RCT intervention studies reported a better socialization quotient (38) and a more favorable behavior (33) in children of iodine-supplemented mothers than in those of mothers who were not supplemented during pregnancy. In the Norwegian Mother and Child Cohort study (MoBa), children born to women with low habitual iodine intake (<160 μg/d) and who took an iodine supplement containing ≤200 μg/d were more likely to have internalizing (but not externalizing) behavioral problems (45), had an increased risk of attention-deficit hyperactivity disorder (ADHD) diagnosis, and had a higher ADHD symptom score at 8 y compared with those of non–supplement users (46).

Discussion

This systematic review has assessed the available evidence on the effects of iodine supplementation in pregnancy in areas of mild-to-moderate iodine deficiency. There was a lack of RCT evidence, and many observational studies were classed as of poor quality. Whereas few studies showed an effect of iodine supplements on maternal or infant TSH or FT_4, most RCTs showed a reduction in maternal Tg, and in half of the RCTs there

TABLE 3 Summary of findings from 14 studies for the effect of iodine supplements during pregnancy on child neurodevelopment[1]

Study design				Cognitive assessment			Child neurodevelopment outcomes
Study, country	Iodine status[2]	Iodine exposure: dose,[3] study group (start of supplementation) (n)	Iodine vehicles	Cognitive test (scales)	Child age (y) at assessment	Assessor	
RCTs ($n = 3$)							
Brucker-Davis et al. (31), France	117 μg/L	A: 150 μg/d, MV + iodine (<10 wk) ($n = 19$) B: MV − iodine ($n = 25$)	MV tablet	Bayley Scales of Infant and Toddler Development—3rd ed.	2 y	Blinded investigator and parent-reported (social-emotional behavior scale)	No differences in composite scores or percentile ranks between groups for any of the outcomes: **Cognitive development:** 110 vs. 110, NS; **Language:** 104.5 vs. 100, NS; **Motor development:** 110 vs. 110, NS; **Social-emotional behaviors score:** 100 vs. 90, NS
Zhou et al. (62), Australia	137 μg/L	A: 150 μg I/d (15.2 wk) ($n = 29$) B: placebo ($n = 30$)	KI tablet	Bayley Scales of Infant and Toddler Development—3rd ed.	1.5 y	Blinded investigator and parent-reported (behavior scales only)	No differences continuously or as percentage with delayed score (score < 85/70) between groups for any of the outcomes: **Cognitive development:** 99.4 vs. 101.7, NS; **Language:** 97.2 vs. 97.9, NS; **Motor development:** 93.9 vs. 92.4, NS; **Social-emotional behavior:** 105.8 vs. 105.4, NS; **Adaptive behaviors score:** 105.2 vs. 103.5, NS
Gowachirapant et al. (34), Thailand and India	131 μg/L	A: 200 μg I/d (10.8 wk) ($n = 412$)[4] B: placebo ($n = 420$)[4]	KI tablet	WPPSI—3rd ed.; BRIEF-P (executive function); SDQ (behavior); Acoustic testing; NBAS (newborn development); BSID—3rd ed.	5.4 y (WPPSI, BRIEF-P, SDQ, Acoustic) 6 wk (NBAS) 1 and 2 y (BSID)	Clinical psychologist administered to the child (WPPSI, NBAS, and BSID) and to the mother of each child (BRIEF-P and SDQ)	No differences continuously or as percentage with delayed score (score < 85) between groups for verbal, performance, and full-scale IQ, processing speed, and global executive function: **Verbal IQ:** 89.5 vs. 90.2, NS; **Performance IQ:** 97.5 vs. 99.1, NS; **Processing speed:** 113.4 vs. 115.0, NS; **Full-scale IQ:** 94.9 vs. 96.1, NS; **Global executive function:** 90.6 vs. 91.5, NS; **Total difficulties:** 9.3 vs. 9.1, NS; **Auditory performance:** left (15.0 vs. 13.3, NS) or right ear (13.3 vs. 13.3, NS); **Newborn neurodevelopment**: no difference overall; **Infant cognitive development**: no difference; **Infant language:** lower expressive language at 1 y in A than in B (14.8 vs. 15.2); **Infant motor development:** no difference

(*Continued*)

TABLE 3 (*Continued*)

Study design				Cognitive assessment			Child neurodevelopment outcomes
Study, country	Iodine status[2]	Iodine exposure: dose,[3] study group (start of supplementation) (n)	Iodine vehicles	Cognitive test (scales)	Child age (y) at assessment	Assessor	
Intervention studies ($n = 3$)							
Berbel et al. (38), Spain[5]	75 μg/L[6]	A: 153 μg I/d (4–6 wk) ($n = 92$)[7] B: 153 μg I/d (12–14 wk) ($n = 102$)[7] C: 153 μg I/d (term) ($n = 151$)[7]	KI tablet (200 μg)	Brunet-Lézine scale, revised 1997 (gross and fine motor coordination, language skills, socialization)	1.5 y	Two blinded specialists	**Total development quotient:** higher in A than in B and C (101.8 vs. 92.2 and 87.5); no difference between B and C (NS); percentage with delayed performance (<85) observed only in B and C (25% and 36.8%); **Gross motor coordination quotient:** higher in A than in B and C (108 vs. 91 and 92)[8]; no difference between B and C (NS); **Fine motor coordination quotient:** higher in A than in B and C (110 vs. 95 vs. 90)[8]; no difference between B and C (NS); **Language quotient:** no difference between groups A, B, and C (96 vs. 92 vs. 90)[8]; **Socialization quotient:** higher in A than in C only (102 vs. 87)[8]; no difference between A or C and B (95, NS)[8]
Velasco et al. (33), Spain	69 μg/L[9]	A: 300 μg I/d (≤13 wk) ($n = 133$) B: no iodine ($n = 61$)	KI tablet (300 μg)	BSID (mental, motor, and behavior scales)	0–1.5 y (mean per group A vs. B: 5.5 vs. 12.4 mo)	Independent researcher blind to the design sequence of the study	**Mental development:** no difference between groups A and B (109.2 vs. 108.9, NS); **Psychomotor development:** higher in A than in B (108.7 vs. 102.7); highest values seen only in breastfed children; **Behavior rating:** higher odds of a more similar or a higher mode than the mode for the age group in A than in B for reaction to persons (OR: 6.93), reaction to the mother (OR: 2.68), cooperation (OR: 22.45), activity (OR: 9.67), arousal (OR: 5.87), and producing sounds (OR: 10.24)
Santiago et al. (32), Spain	109 μg/L	A: 300 μg I/d (<10 wk) ($n = 38$)[10] B: 200 μg I/d (<10 wk) ($n = 55$)[10]	KI tablet (300/200 μg) and IS	BSID (mental and motor scales)	1 y (mean 12.8 mo)	Independent investigator blind to the type of study design	**Mental development:** no difference between groups A, B, and C (104.5 vs. 101.3 vs. 105.6, NS); **Psychomotor development:** no

(*Continued*)

TABLE 3 (*Continued*)

Study design				Cognitive assessment			Child neurodevelopment outcomes
Study, country	Iodine status[2]	Iodine exposure: dose,[3] study group (start of supplementation) (*n*)	Iodine vehicles	Cognitive test (scales)	Child age (y) at assessment	Assessor	
		C: dose n/r, IS (<10 wk) ($n = 38$)[10]					difference between groups A, B, and C (98.6, 94.2, and 100.9, respectively, NS); **Total development**: no difference between groups A, B, and C (203.5, 195.5, and 206.9, respectively, NS) No difference in any of the outcomes when split by long-term IS-use (for ≥1 y prepregnancy)
Prospective cohort studies ($n = 8$)							
Murcia et al. (16), Spain	132 μg/L[11]	A: ≥150 μg/d, iodine-user (≤13 or >13 wk) ($n = 222$) B: 100–149 μg/d, iodine-user (≤13 or >13 wk) ($n = 298$) C: <100 μg/d, iodine-user/nonuser (≤13 or >13 wk) ($n = 169$)	MV tablet	BSID	1 y (mean 12.3 mo)	Trained psychologist	**Mental development:** no difference between groups A, B, and C (99.6 vs. 99.8 vs. 100.7, NS); no association in adjusted analyses between A and C (+0.7 points, NS); no difference in odds of MDI < 85 between A and C (OR: 1.1, NS); **Psychomotor development:** lower in A and B than in C (97.1 vs. 100.6 vs. 102.6); negative association in adjusted analyses between A and C (−5.2 points); higher odds of PDI < 85 in A than in C (OR: 1.8) with stronger association in girls than in boys (OR: 4.0 vs. 1.1)
Rebagliato et al. (17), Spain	125 μg/L	A: ≥150 μg/d, iodine-user (≤13 or >13 wk) ($n = 598$) B: 100–149 μg/d, iodine-user (≤13 or >13 wk) ($n = 228$) C: <100 μg/d, iodine-user/nonuser (≤13 or >13 wk) ($n = 675$)	KI or MV tablet	BSID (mental and motor scales)	1 y (mean 16 mo)	Trained psychologist	**Mental development:** no difference between groups A, B, and C in all subcohorts (Asturias: 96.8 vs. 97.9 vs. 98.6, NS; Gipuzkoa: 98.3 vs. 101.3 vs. 106.7, NS; Sabadell: 99.3 vs. 103.7 vs. 98.3, NS); no association in adjusted pooled analyses between A and C (−1.8 points, NS); no difference in risk of MDI < 85 in adjusted pooled analyses between A and C (OR: 1.7, NS); **Psychomotor development:** no difference between groups A, B, and C in Gipuzkoa and Sabadell (Gipuzkoa: 98.9 vs. 98.9 vs. 99.4, NS; Sabadell: 101.4 vs. 97.4 vs. 99.4, NS); lower in A than in B and C

(*Continued*)

TABLE 3 (*Continued*)

Study design				Cognitive assessment			Child neurodevelopment outcomes
Study, country	Iodine status[2]	Iodine exposure: dose,[3] study group (start of supplementation) (n)	Iodine vehicles	Cognitive test (scales)	Child age (y) at assessment	Assessor	
							in Asturias (Asturias: 93.3 vs. 100.5 vs. 98.2); no association in adjusted pooled analyses between A and C (−0.9 points, NS); no difference in risk of PDI < 85 in adjusted pooled analyses between A and C (OR: 1.5, NS)
Moleti et al. (41), Italy	48 μg/L	A: dose n/r, IS-user (min 24 mo preconception) ($n = 15$) B: no IS-user ($n = 15$) C: dose n/r, IS-user (+LT_4) ($n = 15$) D: dose n/r, no IS-user (+LT_4) (preconception) ($n = 15$)	IS and LT_4	Wechsler Intelligence Scale for Children—3rd ed. (full-scale, verbal, and performance IQ)	6–12 y (mean 9.4–9.8 y)	Trained psychologist (blind)	**Full-scale IQ:** higher in IS-users (A and C) than in nonusers (B and D): A vs. B (93.1 vs. 81.7) and C vs. D (96.1 vs. 81.3); higher by 13 points in IS-users (A + C) than in nonusers (B + D), regardless of LT_4 treatment (94.5 vs. 81.5); 3-fold higher percentage with defective cognitive function in no IS than in IS-users (76.9% vs. 23.1%, OR: 7.7); **Verbal IQ:** higher in IS-users (A and C) than in nonusers (B and D): A vs. B (90.1 vs. 80.3) and C vs. D (97.2 vs. 79.6); higher by 14 points in IS-users (A + C) than in nonusers (B + D), regardless of LT_4 treatment (93.5 vs. 79.6); **Performance IQ:** higher in IS-users (A and C) than in nonusers (B and D): A vs. B (98.2 vs. 87.3) and C vs. D (97.4 vs. 87.5); higher by 10 points in IS-users (A + C) than in nonusers (B + D), regardless of LT_4 treatment (97.4 vs. 87.4)
Murcia et al. (14), Spain	123 μg/L	A: ≥150 μg/d, iodine-user (≤13 or >13 wk) ($n = 610$) B: 100–149 μg/d, iodine-user (≤13 or >13 wk) ($n = 457$) C: <100 μg/d, iodine-user/nonuser (≤13 or >13 wk) ($n = 719$)	KI or MV tablet	McCarthy Scales of Children's Abilities (cognitive and motor scales)	4–5 y (mean 4.8 y)	Trained psychologist	**Cognitive function:** no association with total scale or the subscales in adjusted pooled analyses between A and C (+0.3 points, NS) (mean scores per group not reported); no association of timing or dose of supplement; **Motor function:** no association with total scale or the subscales in adjusted pooled analyses between A and C (+1.2 points, NS) (mean scores per group not reported); no association of timing or dose of supplement

(*Continued*)

TABLE 3 (*Continued*)

Study design				Cognitive assessment			Child neurodevelopment outcomes
Study, country	Iodine status[2]	Iodine exposure: dose,[3] study group (start of supplementation) (n)	Iodine vehicles	Cognitive test (scales)	Child age (y) at assessment	Assessor	
Abel et al. (45), Norway	122 μg/24 h[12]	A: >200 μg/d, iodine-user (0–26 wk preconception ≤22 wk) ($n = 1159$) B: 1–200 μg/d, iodine-user (0–26 wk preconception ≤22 wk) ($n = 14{,}091$) C: nonuser ($n = 33{,}047$)	MV tablet mainly	Language scale ; Ages and Stages Questionnaire (communication and motor skills); Motor milestone (walking unaided); Child behavior checklist	3 y	Mother-reported	In women with habitual iodine intake ≥ 160 μg/d, NS associations with all outcomes; results for women with habitual intake < 160 μg/d reported: **Language delay:** no association in A and B vs. C (OR: 1.02 and 1.06, NS); **Communication delay *z* score:** no association in A and B vs. C ($\beta = 0.04$ and 0.00, NS); **Internalizing behavior problems:** higher odds in B than in C (OR: 1.14) and no difference between A and C (OR: 1.01, NS); **Externalizing behavior problems:** no association in A and B vs. C (OR: 1.21 and 1.07, NS); **Not walking at 17 mo:** no association in adjusted analyses in A and B vs. C (OR: 1.15 and 1.05, NS); **Fine motor delay *z* score:** no association in A and B vs. C ($\beta = 0.00$ and 0.00, NS); **Gross motor delay *z* score:** no association in A and B vs. C ($\beta = 0.02$ and 0.0, NS)
Abel et al. (46), Norway	61 μg/L[13]	A: >200 μg/d (0–26 wk preconception ≤22 wk) ($n = 1864$) B: 1–200 μg/d (0–26 wk preconception ≤22 wk) ($n = 21{,}940$) C: nonuser ($n = 53{,}360$)	MV tablet mainly	ADHD diagnosis; ADHD Rating Scale (ADHD symptom score)	8 y	Patient registry (diagnosis) and mother-reported (symptoms)	**ADHD diagnosis:** no difference in risk between A and B vs. C; increased risk in B vs. C only when supplement started at 0–12 wk in women with low habitual iodine intake < 160 μg/d (HR: 1.29); **ADHD symptom score:** higher in B than in C in some adjusted analyses ($\beta = 0.05$) and in fully adjusted analyses only when supplement started at 0–12 wk ($\beta = 0.06$) in women with iodine intake < 160 μg/d

(*Continued*)

TABLE 3 (*Continued*)

Study design				Cognitive assessment			Child neurodevelopment outcomes
Study, country	Iodine status[2]	Iodine exposure: dose,[3] study group (start of supplementation) (n)	Iodine vehicles	Cognitive test (scales)	Child age (y) at assessment	Assessor	
Abel et al. (48), Norway	67 μg/L[14] (91 μg/g)	A: dose n/r, iodine-user (0–26 wk preconception ≤2 wk) ($n = 14{,}665$) B: nonuser ($n = 24{,}806$)	MV tablet mainly	The Children's Communication Checklist-Short (CCC-S) (language skills); Vineland Adaptive Behaviour Scale-II: (selected 3 questions on reading and 2 on writing skills); mapping tests (reading and maths); special education in school	8 y	Mother-reported (language, reading, and writing skills; reading and maths mapping tests; granted special education services)	**Language skills:** NNSn adjusted analyses ($\beta = 0.00$); **Reading skills:** NS association in adjusted analyses ($\beta = -0.01$); **Writing skills:** NS association in adjusted analyses ($\beta = -0.01$); **Reading mapping test:** no difference in the odds of attaining poor test results (OR: 0.98); **Maths mapping test:** no difference in the odds of poor test results (OR: 0.99); **Special education:** no difference in the likelihood of receiving special education services (OR: 0.96); NS associations with the outcomes when using timing of supplementation as the exposure variable (nonusers, prepregnancy, 0–12 wk, or > 12 wk)
Markhus et al. (15), Norway	78 μg/L (82 μg/g)	A: 150–250 μg/d, iodine-user (≤ mean 23.7 wk) ($n = 155$) B: nonuser ($n = 658$)	MV	Bayley Scales of Infant and Toddler Development—3rd ed. (cognitive, language, and motor scales)	Mean 6.1 and 12.2 mo (screening version); mean 18.4 mo (full-scale version)	Trained health care nurses	**Cognitive score:** NS association in adjusted analyses ($\beta = 0.02$); **Receptive language:** NS association in adjusted analyses ($\beta = -0.02$); **Expressive language:** NS association in adjusted analyses ($\beta = 0.05$); **Fine motor skills**: NS association in adjusted analyses ($\beta = -0.10$); **Gross motor skills:** significantly lower in A than in B in adjusted analyses ($\beta = -0.18$)

[1] ADHD, attention-deficit hyperactivity disorder; BRIEF-P, Behaviour Rating Inventory of Executive Function—Preschool Version; BSID, Bayley Scales of Infant Development; IQ, intelligence quotient; IS, iodized salt; KI, potassium iodide; LT_4, levothyroxine; MDI, Mental Development Index; MV, multivitamin/mineral supplement; NBAS, Neonatal Behavioural Assessment Scale; n/r, not reported; NS, not statistically significant; PDI, Psychomotor Development Index; RCT, randomized controlled trial; SDQ, Strengths and Difficulties Questionnaire; UIC, urinary iodine concentration; UIE, urinary iodine excretion; WPPSI, Wechsler Preschool and Primary Scale of Intelligence.

[2] Baseline iodine status was expressed as median UIC (μg/L), median urinary iodine-to-creatinine ratio (μg/g), or median UIE (μg/24 h) of the total sample. If baseline iodine status values were not available for the total sample, values for the control group/group who did not use iodine-containing supplements and/or IS were used.

[3] In some cases where the authors only provided the KI dose used but not the actual iodine content of the tablets, the dose was converted to micrograms of iodide to facilitate comparisons between studies.

[4] The numbers of children with cognitive tests were smaller (e.g., for WPPSI, A = 159 compared with B = 154 and for BRIEF-P, A = 159 compared with B = 156) in Gowachirapant et al. (34).

[5] Women in this study were a very select group (i.e., A = euthyroid compared with B and C = hypothyroxinemic in first trimester or at term, respectively) in Berbel et al. (38).

[6] Median UIC values at baseline were not reported and values measured at delivery/term were used for this study in Berbel et al. (38).

[7] The numbers of children with cognitive tests were smaller (e.g., A = 13, B = 12, and C = 19) in Berbel et al. (38).

[8] Values were estimated from bar charts in Berbel et al. (38).

[9] Baseline iodine status value was based on a subgroup of women from the area who had miscarriage and were not included in this study in Velasco et al. (33).

[10] The numbers of children with cognitive tests were smaller (e.g., A = 30, B = 47, and C = 25) in Santiago et al. (32).

[11] Baseline iodine status value was extracted from Rebagliato et al. (35), because the median UIC of the exact sample was not reported in this study in Murcia et al. (16).

[12] Baseline iodine status value for this study was based on 24-h UIE and iodine intake estimated from an FFQ in Abel et al. (45).

[13] Baseline iodine status was measured only in a subset of 1950 women for this study and the presented total value is based only on women who did not use iodine-containing supplements in Abel et al. (46).

[14] Baseline iodine status was measured only in a subset of 2001 women at mean 18.5 wk in Abel et al. (48).

A Cognitive scores

Study (ref.)	Iodine Total *n*	Iodine Mean	Iodine SD	Placebo Total *n*	Placebo Mean	Placebo SD	Weight	MD (95% CI)
Zhou et al. (62)	27	99.4	12.2	26	101.7	8.2	3.5%	-2.30 (-7.88, 3.28)
Gowachirapant et al. (34)	208	63.0	5.3	202	63.1	5.7	96.5%	-0.10 (-1.17, 0.97)
Random-effects model	**235**			**228**			**100%**	**-0.18 (-1.22, 0.87)**

Heterogeneity: $I^2 = 0\%$, $\tau^2 = 0$, $P = 0.45$

B Language scores

Study (ref.)	Iodine Total *n*	Iodine Mean	Iodine SD	Placebo Total *n*	Placebo Mean	Placebo SD	Weight	MD (95% CI)
Zhou et al. (62)	27	97.2	12.2	26	97.9	11.5	5.9%	-0.70 (-7.08, 5.68)
Gowachirapant et al. (34)	203	54.8	7.9	196	53.4	8.4	94.1%	1.40 (-0.20, 3.00)
Random-effects model	**230**			**222**			**100%**	**1.28 (-0.28, 2.83)**

Heterogeneity: $I^2 = 0\%$, $\tau^2 = 0$, $P = 0.53$

C Motor scores

Study (ref.)	Iodine Total *n*	Iodine Mean	Iodine SD	Placebo Total *n*	Placebo Mean	Placebo SD	Weight	MD (95% CI)
Zhou et al. (62)	27	93.9	10.8	26	92.4	9.7	6.2%	1.50 (-4.02, 7.02)
Gowachirapant et al. (34)	202	96.0	7.7	199	95.8	6.8	93.8%	0.20 (-1.22, 1.62)
Random-effects model	**229**			**225**			**100%**	**0.28 (-1.10, 1.66)**

Heterogeneity: $I^2 = 0\%$, $\tau^2 = 0$, $P = 0.66$

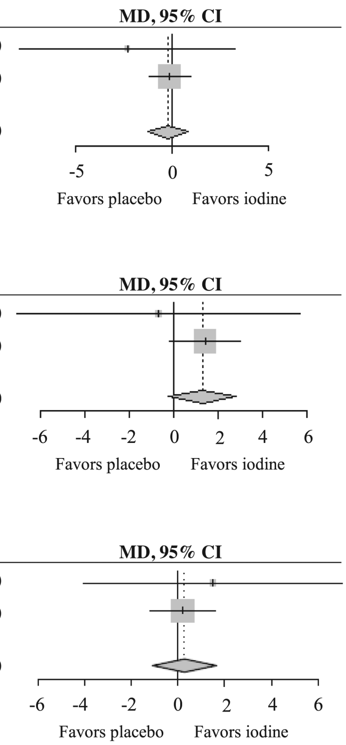

FIGURE 2 Forest plots of the effect of maternal iodine supplementation compared with placebo on child cognitive (A), language (B), and motor (C) scores. Data were analyzed using random-effects meta-analysis. MD, mean difference; ref., reference.

was a lower increase in maternal thyroid volume with advancing gestation. Only a few studies showed lower cord-blood Tg and lower infant thyroid volume with iodine supplementation.

Overall there was a lack of data on child cognitive outcomes with only 1 adequately powered RCT (34); that RCT showed no effect on child motor or mental development, or behavior. Results from cohort studies and non-RCT interventions showed no effect of maternal iodine supplements on mental development (14–17, 32, 33, 38, 45, 48), but there were inconsistent effects on motor development, both harmful (15–17) and beneficial (33, 38), respectively. Non-RCT interventions showed positive effects on externalizing-type behavior (33, 38), whereas cohort studies reported negative associations with internalizing behavior (45) and ADHD symptoms/diagnosis (46) in those with low iodine intake (although after adjustment for multiple testing and using matched controls, these effects were attenuated).

We suggest that the inconsistencies in the evidence in mild-to-moderate iodine deficiency might be explained by *1*) maternal prepregnancy/baseline iodine status; *2*) dose and form of iodine supplement; *3*) timing of supplementation; and *4*) the cognitive tests used.

Firstly, maternal intrathyroidal iodine stores accumulated prepregnancy (4) may allow the thyroid to adapt to low iodine intake in pregnancy and maintain normal function despite the increased demands of pregnancy. However, adaptation requires enhanced thyroidal stimulation (4) which may put a strain on the maternal thyroid resulting in an elevation of serum Tg and, progressively, goitrogenesis. Indeed in most studies, iodine supplementation resulted in a reduction in Tg, although not maternal thyroid volume which may take longer to respond (65). Because the early stages of pregnancy are a critical time for adequate iodine supply (66), supplementation after confirmation of pregnancy might be too late. Indeed, a series of cohort studies (39–41) has shown that mothers who used iodized salt before pregnancy, thus likely optimizing their iodine stores, had lower TSH and higher FT_4 during pregnancy than those who either did not use iodized salt, or started using it in pregnancy. Prepregnancy iodine stores might have affected the null outcome in the Gowachirapant et al. (34) RCT because it was set in countries that had universal salt iodization.

An abrupt increase in iodine intake during pregnancy might have negative effects, especially in those with low prepregnancy iodine intake. Women who started taking iodine supplements (40) or iodized salt (39) on becoming pregnant had higher TSH and lower FT_4 than women who had been using iodized salt prepregnancy. Another study showed that the introduction of iodine-containing supplements at 13 wk was associated with lower FT_4, whereas longer-term use was not (47). In women with a habitually low iodine intake (<160 μg/d), iodine supplements were negatively associated with child behavior, including an increased risk of ADHD diagnosis, higher ADHD symptom-score, and increased odds of a high score for internalizing-behavior problems (45, 46).

Differences in maternal baseline iodine status might also partly explain the inconsistent results on thyroid function. For instance, iodine supplements lowered cord-blood Tg only in the studies where baseline median UIC was <100 μg/L (49, 50) but not in the studies where median UIC was >100 μg/L (31, 62). Similarly, iodine supplements resulted in lower infant thyroid volume in studies with a median UIC <100 μg/L (53, 54) but not in 1 study with a median UIC of 109 μg/L (32). In observational studies where baseline UIC was >100 μg/L (35, 37, 59), mothers who used iodine supplements during pregnancy had higher TSH and/or there was a higher percentage with elevated TSH. The RCTs with baseline UIC >100 μg/L found mostly no difference in maternal TSH or FT_4 between the iodine and control groups (31, 34, 52); by contrast, in a number of RCTs with median UIC <100 μg/L (44, 50, 51, 61) iodine supplementation prevented or reduced the increase in TSH over the course of pregnancy.

Secondly, form and dose of iodine may affect results. In 16 studies iodine was part of a multivitamin/mineral supplement, making it hard to isolate its effect from that of other components. In the observational studies that showed negative effects of iodine supplementation on infant psychomotor development (dosage ≥150 μg/d) (15–17) and behavior (dosage ≤200 μg/d) (45, 46), iodine was supplied mostly as a multivitamin/mineral preparation. Sometimes it was unclear whether the dose was expressed as μg KI or μg iodide (from KI); because only 76% of KI is iodide (67), the dose must be clearly specified. None of the studies that looked only at specific KI supplements reported negative effects on child neurodevelopment. Regardless of supplement type, in observational studies it is important to consider the behaviors associated with supplement taking that could bias results in either direction (e.g., worried/health-seeking behavior of the mother) (46).

Thirdly, the timing of supplementation is also likely to be important. An intervention study administering iodine supplements at different times during pregnancy found that supplementation at 4–6 wk was more effective in improving infant neurodevelopment than supplementation at 12–14 wk or later (38). In that study, however, women in the group with the earliest administration of supplement were euthyroid at baseline, whereas the other 2 groups with later administration were hypothyroxinemic, thus the effect of timing of iodine supplementation cannot be distinguished from that of maternal baseline FT_4 concentrations (38). Two RCTs (31, 34) and 1 non-RCT intervention study (32) that administered iodine in the first trimester (∼10 wk) showed no effect on child development, whereas 1 non-RCT intervention study (33) reported a beneficial effect. Although all these studies administered iodine early in pregnancy, they showed mixed results which may relate to prepregnancy status, or a lag period before any benefits of iodine supplementation are seen (4).

Finally, the methods for assessing child neurodevelopment may explain null findings. In most studies, global assessments [e.g., Bayley Scales of Infant Development (BSID)] were used which, if conducted in infancy, may have low predictive capacity for childhood intelligence and behavior (68, 69). Indeed, in INMA, the negative effect on motor development (using BSID) at 1 y (16, 17) was not confirmed at age 4–5 y (14). Although cognitive tests may be more valid at older ages, there are also more confounders (70, 71). Understanding specific brain systems that are vulnerable to iodine deficiency might enable iodine-sensitive cognitive tasks to be used in future trials, e.g., visual information-processing (72).

There was some weak evidence, mainly from observational studies, that supplement use was associated with elevated maternal (35, 37, 59) and newborn TSH (33, 49), and some negative effects on child psychomotor development (15–17) and behavior (45, 46). We found no evidence that iodine supplements were associated with markers of maternal or infant thyroid-autoimmunity. Negative effects were not confirmed in an RCT [with 200 μg I/d (34)] but further evidence of safety is needed in areas of moderate deficiency.

Our study is limited by the fact that we only included articles in English and that we used nonvalidated cutoffs (UIC of 50–149 μg/L) to identify mild-to-moderate deficiency in pregnancy (26), thus study misclassification may have occurred.

In summary, there is insufficient good-quality evidence to support current recommendations for iodine supplementation during pregnancy in areas of mild-to-moderate iodine deficiency. There is a need for well-designed randomized trials with child cognitive outcomes in pregnant women who are moderately iodine deficient (i.e., median UIC <100 μg/L). Considering the potential importance of iodine stores, future trials should include appropriate measures (e.g., dietary questionnaire) of prepregnancy iodine intake as part of the study design. However, ethical and feasibility considerations are likely to limit the potential for future iodine RCTs.

The authors' responsibilities were as follows—MPR and SCB: devised the study; MD, HF, and SCB: screened the abstracts; MD and HF: extracted the data from the selected studies; JM: performed the meta-analyses; MD: wrote the first draft of the manuscript; and all authors: contributed to the manuscript and read and approved the final manuscript. The authors report no conflicts of interest.

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